



AmelioSense™

Calm the flame of pyroptosis – soothe irritated skin



AmelioSense™ is a comprehensive approach aimed at effectively combating skin irritation and redness. AmelioSense™ combines the extract of shepherd's purse with a powerful liposomal antioxidant complex. AmelioSense™ works effectively against pyroptosis, a programmed, highly inflammatory cell death, while at the same time reducing general skin inflammation processes that lead to redness and irritated skin.

The issue

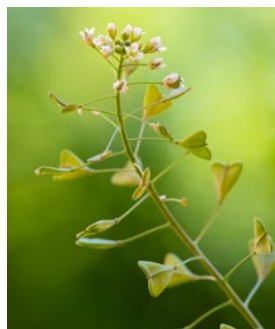
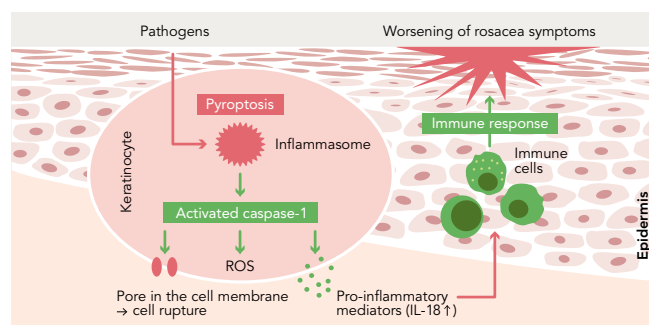
Rosacea is a chronic inflammatory skin condition that affects millions of people worldwide, causing redness, visible blood vessels, and discomfort primarily on the face. Despite its prevalence, the exact causes of rosacea remain elusive, making effective treatment a challenge. In addition to rosacea, many people experience skin irritation, flushing and redness with underlying inflammation processes.

The solution

Recent research has unveiled a critical link, which shows that pyroptosis can exacerbate rosacea. The activation of the inflammasome and the subsequent pyroptotic processes are central to the chronic inflammation characteristic of rosacea. The pro-inflammatory mediators that are released during pyroptosis promote the infiltration, activation, and differentiation of immune cells, which heightens the immune and inflammatory response and worsens rosacea symptoms. Therefore, targeting pyroptosis is essential for optimizing care of rosacea and inflammation.

The benefit

AmelioSense™ is an innovative new approach to effectively fight pyroptosis, irritation, and skin redness. *In vitro* and *in vivo* studies have demonstrated that AmelioSense™ inhibits active caspase-1, the key enzyme in pyroptosis, inhibits the gene expression of diverse inflammatory markers known to be also present in rosacea, and visibly reduces the appearance of blood vessels and redness in the cheek area.



Source

Shepherd's purse, a plant from the cruciferous family (*Brassicaceae*), has a long history in natural medicine, particularly for its anti-inflammatory properties and ability to promote wound healing. Its scientific name, *Capsella bursa-pastoris*, refers to the shape of its fruits, which resemble the pockets of shepherds from ancient times. AmelioSense™ combines shepherd's purse extract with an advanced antioxidant complex that includes carnosine, tocopherol, and silymarin.

INCI (EU/PCPC) Declaration: Capsella Bursa-Pastoris Extract (and) Lecithin (and) Carnosine (and) Tocopherol (and) Silybum Marianum Fruit Extract (and) Maltodextrin (and) Aqua/Water



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Study results

Inhibition of the Pyroptosis Key Enzyme Caspase-1

S-889



Cell type: Normal human epidermal keratinocytes

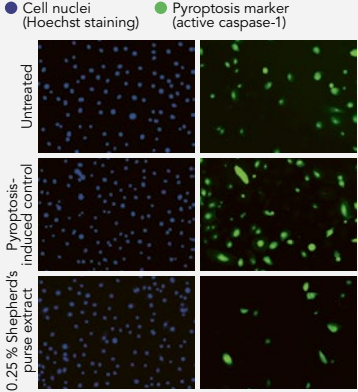
Test compound/ treatment: Cells were pretreated for a period of 24 hours with 0.25 % shepherd's purse extract or not (control). The medium was then replaced by medium containing HMGB1, a well-known pyroptosis trigger, either alone or together with the extract for a further 24 hours

Parameter: Active caspase-1 (fluorescence microscopy)

→ Shepherd's purse has a significant inhibitory effect on pyroptosis by reducing active caspase-1 as the key enzyme in the pyroptosis pathway

→ AmelioSense™ could stop the spreading of the inflammation and thus has the potential to soothe irritated skin

● Cell nuclei (Hoechst staining) ● Pyroptosis marker (active caspase-1)



Visible Improvement of Skin Redness

S-1211



Volunteers: 22 (f/m, 29–70y) with redness, irritated skin and visible blood vessels in the cheek area

Test compound/ treatment: Application of a cream containing 2 % AmelioSense™ on one side of the face twice a day, placebo to the other side

Parameter: High-resolution images (ColorFace® system)

→ AmelioSense™ had a significant impact in reducing visible skin redness, flushes and irritations in the face



Reduction in Redness and Blood Flow

S-1211

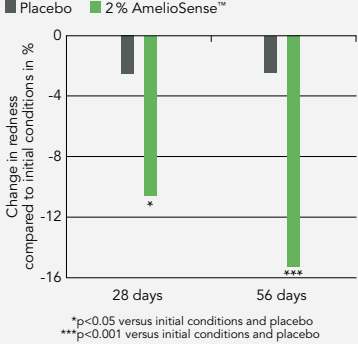


Volunteers: 22 (f/m, 29–70y) with redness, irritated skin and visible blood vessels in the cheek area

Test compound/ treatment: Application of a cream containing 2 % AmelioSense™ on one side of the face twice a day, placebo to the other side

Parameters: Skin redness (Spectrophotometer® CM-700d), blood flow in capillaries (Laser Doppler VMS-LDF2® instrument)

→ Redness was significantly reduced by 15 % and blood flow by 46 % after 56 days of treatment with 2 % AmelioSense™



Time Point	Placebo (Change in redness %)	2 % AmelioSense™ (Change in redness %)
28 days	~0	~-10.5*
56 days	~-2.5	~-14.5***

*p<0.05 versus initial conditions and placebo
***p<0.001 versus initial conditions and placebo

In Addition

S-1060/S-889



Reduction of Inflammatory Markers: The antioxidant complex of AmelioSense™ was applied to keratinocytes in a rosacea-like environment. The complex significantly reduced key inflammatory markers, such as CXCL1, CXCL2, IL8, IL23A, and TNF-alpha by up to 95 %. These findings demonstrate AmelioSense™'s potent anti-inflammatory properties.



Reduction in ROS-Release and IL-18 Gene Expression: Pretreating human epidermal keratinocytes with 0.25 % shepherd's purse extract significantly reduced both, the increase in reactive oxygen species (ROS) and the IL-18 gene expression caused by HMGB1-induced pyroptosis, bringing them back to normal levels. Therefore, AmelioSense™ could strongly contribute to combating irritation and redness in the skin.

Recommended use level: 2 %

Characteristics: Powder, water-soluble, without preservatives and alcohol

Applications: Soothing anti-redness serum for delicate skin, redness relief soothing cream, ultra calming anti-redness mask, instant relief barrier strengthening serum, redness relief daily cleansing milk

Benefits AmelioSense™

- Reduces the appearance of visible blood vessels on the cheeks
- Decreases facial redness and blood flow
- Alleviates irritation
- Promotes a calm and balanced skin tone